

WUDA: Visualizing and Transforming Rotations in Real-Time with Quaternions and Smart Devices

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ABSTRACT

The rising popularity of inertial sensing via smart devices is evident, finding use in multidisciplinary applications, such as gesture recognition and image stabilization. Amid the popularity of analyzing raw motion signals and Euler angles, we highlight quaternion rotations as a robust alternative for studying device orientation. Quaternions are a powerful mathematical tool for representing and affecting three-dimensional rotations. Their abstract nature, however, can make it difficult for researchers to visualize the data they provide. Mobile sensing rarely invents practical tools to experiment with this mathematical method. Since visualization effectively communicates data findings, we develop an open-source, real-time app, Wuda, that allows users to observe the orientation of smart devices via inertial sensing. Wuda helps users reduce dimensional complexity and intuitively study quaternion transformations. We demonstrate the practical nature of Wuda in the context of fitness tracking.

Keywords: Mathematical Foundations & Numerical Methods, Machine Learning, Statistics, Modelling, and Simulation Applications

Index Terms: Human-centered computing—Visualization—

1 INTRODUCTION

With the widespread commercialization of wearable devices, inertial sensing has become an affordable mechanism for studying system behavior. The technology has found its way into various industries with applications in health, navigation, sports analysis, and activity recognition. However, most off-the-shelf devices lack complementary software to help users visualize the movement of smart devices in space. Thus, the limitation presents a conundrum wherein researchers seeking to understand and apply this form of sensing must gain a comprehensive understanding without visual aid. Perhaps, the intricate mathematical nature develops the dilemma, as there is no easy mechanism to convey and articulate the nuances of rotation.

Modern devices, such as smartphones, leave exploration to the user by exposing various mathematical methods, including Euler angles, Rotation matrices, and Quaternions. Euler angles describe trajectories as a product of three rotations about three axes. Tait–Bryan angles use three distinct axes, and proper Euler angles use the same axis for the first and last element, e.g., x, y, z or x, y, x respectively [6]. Given the subtle differences, we refer to both representations as Euler angles throughout the text. Despite their ease of implementation, the resulting trajectory changes based on the order in which users apply transformations and are susceptible to singularities at the poles [8]. Contrary to user expectations, this technique can sometimes cause a dramatic change to, what seems to be, a slight rotation. On the other hand, rotation matrices and quaternions do not have the same limitations. Instead, they provide finer predictability at the cost of increasing complexity.

A more natural interpretation of the two mathematical methods is rotation matrices, which live in three-dimensional space. Rotation matrices have an intuitive representation of dimensions, mak-

ing them straightforward for those familiar with matrix operations. While matrix representations of rotations may appear advantageous at first, they are subject to numerical drift and inefficiency in terms of memory utilization. Conversely, quaternions live in four dimensions. Aside from complexity and dual covering, their compact algebraic structure, interpolation properties, and ability to affect rotations about an arbitrary axis make them a superior alternative to other techniques. Unfortunately, quaternion rotations are often an advanced topic in mathematics due to the difficulty in portraying four-dimensional space [14, 16, 24, 28].

In this work, for the reasons given above, we leverage quaternions as a mechanism to visually describe orientation. Specifically, we focus on unit-norm quaternions and their natural application to rotations with smart devices. Today, one must use commercial and, at times, advanced software, such as MATLAB or Unity [15, 25]. These programs have specialized algorithms to convert quaternions into a visual representation. However, interacting with available tools can be complex, requiring specific technical and mathematical knowledge. For instance, designing and rotating an object in Unity is trivial with the built-in tools. The task can become more challenging when working with inertial sensors as it requires understanding how to integrate and manipulate the data to display the desired information correctly [1]. This phenomenon creates a circular dependency, where comprehending quaternions necessitates their visualization, and conversely, visualizing them necessitates a prior understanding of quaternions. We aim to reduce the entry barrier to inertial sensing and rotations and present the following salient contributions in this work:

- Develop a native open-source macOS application, Wuda, that allows users to experiment with smart sensing without an explicit need for commercial software.
- Design a real-time interface that interprets rotations in three, two, and one-dimensional space.
- Provide a real-time control panel that allows users to transform motion signals.

2 MOTIVATION

Despite the popularity of commercial and open-source graphing tools, interactive visualization of rotations is limited. It is only with advanced cutting-edge technology, such as motion capture systems [10], that individuals can quickly realize motion. Because of this imbalance, individuals often resort to a sequential process that starts with data collection, then tool exploration, and eventually the study of gathered material. The natural tendency to repeat this cycle decouples the researcher's ability to collect and process information simultaneously and increases the cognitive load to recall previous events. This limitation intensifies with dynamic systems which abide by external constraints, such as schedules and the occurrence of measurable states. Ultimately, prototyping algorithms and intuitively understanding the behavior of systems prolong as the visualization occurs after an experience naturally unfolds.

3 SYSTEM DESIGN AND OVERVIEW

In this section, we discuss the properties of quaternions and how our visualization tool enables easier comprehension of rotations.

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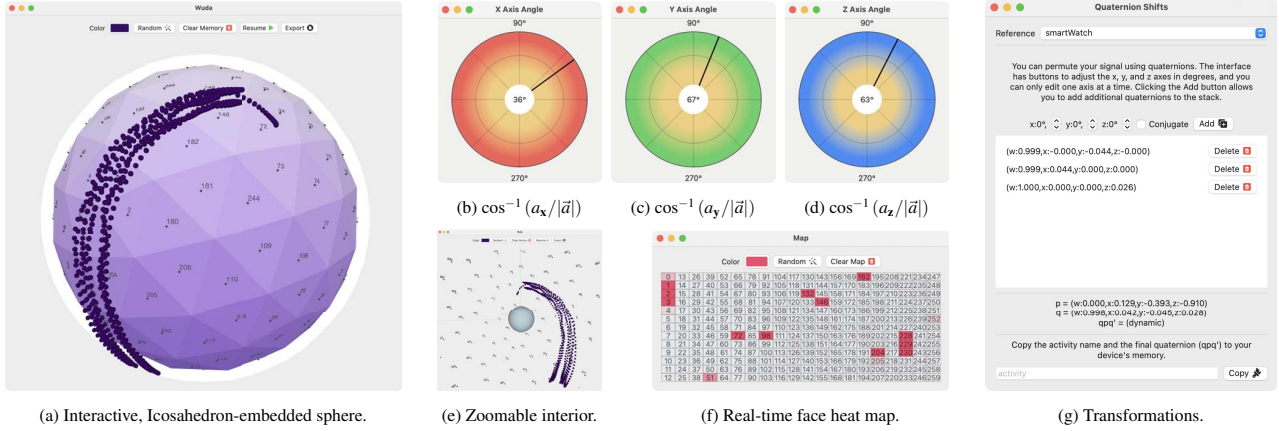


Figure 1: Wuda draws an interactive sphere visualization that embeds a Geodesic Icosahedron with labeled faces. In real-time, it rotates a vector p with a quaternion q , $a = qpq^{-1}$. The interface plots a onto the sphere, Figures 1a, 1e, updates the angle between a and x -axis, y -axis, and z -axis, Figures 1b, 1c, 1d, and refreshes the face heat map by determining the closest face to a , Figure 1f. Users transform the signal using the control panel in Figure 1g.

3.1 Overview

Wuda is a standalone open-source MacOS application that allows users to visualize and permute motion signals¹. Wuda receives data over a Bluetooth service it broadcasts to subscribing clients. To reduce the entry barrier to comprehending rotations, we build a complementary standalone Apple Watch app, Wudica, that subscribes to Wuda's services. Note that any system that supports BLE can interact with Wuda. Wudica captures device motion via the HealthKit framework and transfers data to Wuda in the form of unit-norm quaternions.

3.2 Primer on Rotations

We define a quaternion, q , and its conjugate, q^* , as an ordered pair containing a scalar, s , and a vector part, $\mathbf{v}$. Unit-norm and pure quaternions are special terms, where the former is normalized such that $s^2 + \mathbf{v}^2 = 1$ and the latter has a scalar component of 0. If we express $\mathbf{v}$ in terms of its members, i, j and k represent the imaginary parts that satisfy the rule $i^2 = j^2 = k^2 = ijk = -1$. Consult Equation 1 for a formal definition and additional properties of unit-norm quaternions. Notice the simplicity of constructing the conjugate or the inverse.

$$\begin{aligned} q &= [s, \mathbf{v}], \quad q^{-1} = [s, -\mathbf{v}], \quad q^{-1} = q^* \\ q &= [s, x\mathbf{i} + y\mathbf{j} + z\mathbf{k}] \quad s, v, x, y, z \in R \end{aligned} \quad (1)$$

Quaternions help realize and affect how a vector, p , rotates in space. Encoding the vector p as a pure quaternion, $p = [0, p]$, the product $q_1 p q_1^{-1}$ rotates the vector counter-clockwise based on the axis and angle, θ , of rotation defined by a unit-norm quaternion q_1 . This technique extends to motion analysis, computer animation, aircraft control systems, robotics, etc. It's worth noting that the leading q_1 and trailing q_1^{-1} rotation is necessary to align p with the desired frame of reference. The order of the two components dictates direction, where $q_1^{-1} p q_1$ changes the rotation to clockwise [13, 27]. Algebraic properties of unit-norm quaternions also enable an efficient mechanism for the multi-composition of transformations. A vector p is subject to a second rotation, q_2 , in the following expression: $(q_2)(q_1)(p)(q_1^{-1})(q_2^{-1})$. Strategic compositions can alter the trajectories and patterns that inertial sensors generate. Note that double-angle trigonometric terms can emerge when one applies rigid

transformations to p [27]. Applying a $\theta = 45^\circ$ around the x -axis produces an unintentional 90° rotation. For this reason, it is necessary to halve θ to intentionally rotate the vector by θ , as seen in Equation 2 [27].

$$q = [\cos(\frac{\theta}{2}), \sin(\frac{\theta}{2})\mathbf{v}] \quad (2)$$

3.3 System Design

Wuda is composed of several interactive windows that update in real-time. The main window is an interactive three-dimensional scene containing a unit sphere that embeds an irregular polyhedron. Each face of the polyhedra, a triangular shape in our application, includes a text label and a center point. When subscribing clients transmit data to Wuda, they specify p and q . p can take a dynamic form, representing a frame of different sensors in the smart device, or a static form, such as the unit vector in the direction of the z -axis. q represents the rotation of the smart device. Wuda rotates the vector using the product qpq^{-1} , and the result, a , populates the sphere's surface.

Labels on the polyhedra serve as a localization mechanism for the user and enable Wuda to find the closest face for each a . This context allows users to reduce quaternion rotations to two dimensions using heat maps, as seen in Figure 1f. We color code the sphere to allow an association to poles and delegate color selection of a to the user. Figure 1a demonstrates a Geodesic Icosahedron with 260 faces and a dark-purple to a light-purple color gradient. Figure 1f contains the associated heat map. Since inertial sensing is multidisciplinary and visualization precision can differ, we provide additional irregular polyhedra with 60, 80, 180, 380, and 780 faces. We borrow the design for objects using an interactive online resource². Since Wuda is open-source, users can opt out to introduce even more granularity to the mesh using custom designs. Zooming into the sphere allows users to visualize the trajectory from the interior and further analyze rotations using the illuminated world axis, as seen in Figure 1e.

The final visualization provided by Wuda reduces rotations to one dimension by calculating the angle between a and the x -axis, y -axis, and z -axis. We calculate the angle by using the property of dot products, $\vec{a} \cdot \vec{i} = |\vec{a}||\vec{i}| \cos \theta$, where $\vec{i}$ is a unit vector along an axis. A dial moves on the unit circle to indicate the current position in degrees. Figures 1b, 1c, and 1d demonstrate the latest angle between

¹<https://github.com/smilanko/wuda>

²<http://dmccoey.com/polyhedra/index.html>

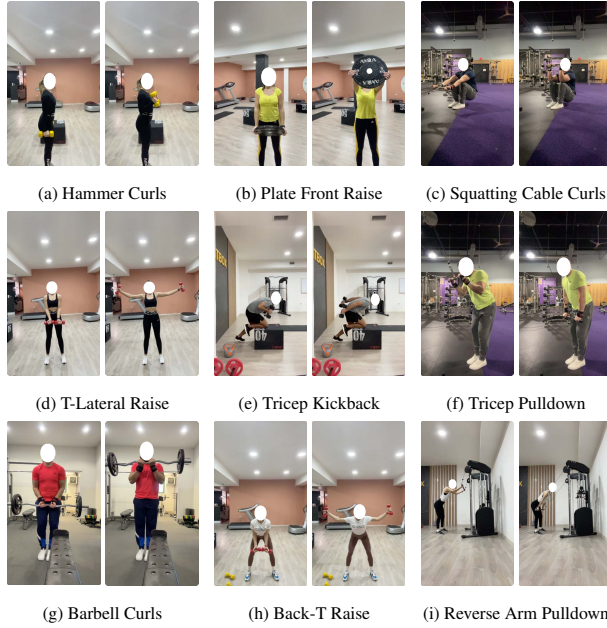


Figure 2: Volunteers performing free-weight exercises. Image pairs demonstrate the starting and ending position of a repetition.

a and the three axes. This graphic is especially advantageous in human kinematics for estimating joint movement.

3.4 Transformations

Wuda allows users to chain transformations using a control panel. The interface in Figure 1g presents a selector in which users toggle an axis and the amount of rotation. We use this as input for Equation 2. Since many applications to inertial sensing deal with human perception and interpretation of angles, Wuda infers units in degrees. Users can add and remove any transform and observe the result in real-time.

4 CASE STUDY

In this section, we demonstrate the application of Wuda to exercise physiology. We discuss popular weightlifting exercises, the trajectories human kinematics can generate, and how Wuda helps apply additional transformations to influence the associated patterns. It's important to note that Wuda and quaternion's algebraic properties apply to a wide range of use cases. Ideas and methods in this paper are not limited to a particular domain, activity, or gesture and have the potential for various applications.

4.1 Visualizing

We ask seven volunteers to complete a favorite free-weight exercise. All participants wear an Apple Watch 7 on their left wrist and toggle the beginning and end of a set using the Wudica app. We set p to the unit vector in the direction of the z -axis, and q to the device attitude using the $xArbitraryCorrectedZVertical$ reference frame [3]. We demonstrate all the selected exercises in Figure 2 and the associated digital representation, qpq^{-1} , in Figure 3.

4.1.1 Exercise Execution

Squatting Cable Curls, Tricep Pulldowns, Barbell Curls, and Reverse Arm Pulldowns are isolated exercises that target different muscle groups. Trainees often engage in various postures, starting limb positions, and directions of movement to complete a single



Figure 3: For each exercise, Wuda generates different patterns on the unit sphere.

repetition. Despite these differences, motion trajectories in Figures 3c, 3f, 3g, and 3i are nearly coincident. In our experiment, this group of exercises tends to encompass the entire y -axis of the sphere. An inverse observation exists for Plate Front Raise and Tricep Kickback, Figures 3b and 3e, which form a unique group of their own. The benefit of visualizing workouts using Wuda is categorizing exercises that resemble a similar digital representation. This context is conducive when building clustering and activity detection algorithms. Wuda also helps identify the associated range of motion of performing repetitions and which axis the exercise rotation targets. For instance, the upper limbs travel similarly during T-Lateral Raise, Figure 3d, and Plate Front Raise, Figure 3b, exercises. However, the wrist is oriented differently during execution and, thus, naturally generates a different pattern.

4.1.2 Heatmap

A current challenge in Smart Health is detecting workout repetitions before their completion [18]. The goal is to provide instantaneous feedback to the trainee via a system that can detect exercise motion as it is happening. This capability has the potential to guide users in safely executing their exercises. In order to perform early localization of reps, the solution needs accurate filtration techniques to remove noise from the kinematic signal, such as trainees stretching or hydrating between sets. Noise can take many forms, and focusing on the patterns generated by a specific activity is often more insightful, as these patterns exhibit greater stability and consistency than random human movement. To understand where exercise motion lives on the geodesic icosahedron, i.e., what is not noise, we observe the real-time heatmap of a single Tricep Pulldown set. We leverage a mesh with 180 and 80 faces and demonstrate the final result in Figures 4a and 4b, respectively. Workout motion occupies approximately 13% of the larger icosahedron and 26% of the smaller. This finding suggests that reducing the mesh size can introduce unwanted overlap between exercise motion and noise. Further reducing the number of faces from 80 to 20 in this experiment presents a segmentation failure where the trainee's workout behavior is indistinguishable from arbitrary activities.



Figure 4: We monitor the heatmap of the exercise motion for a single Tricep Pulldown set. We change the mesh size of the Geodesic Icosahedron to determine the relationship to the spread of data, as shown in Figures 4a and 4b.

4.2 Transforming

In prior work, researchers show that strength training repetitions resemble a U-shaped pattern. A proper rep consists of two phases, concentric and eccentric, in which trainees lift or lower the weights [17]. Thus, the movement's digital trail would have a negative slope in the first phase and a positive in the other. By leveraging peak detection algorithms, they detect reps with high accuracy. To determine if an exercise in our experiment corresponds to the signal and qualifies as input to the algorithm, we select a single Barbell Curls repetition and observe the angle between the resulting rotation and the z -axis. To start this exercise, a trainee holds the weights before their body. The forearm then rotates towards the chin, causing the bicep muscle to contract. Figures 5a-5c demonstrate the angle between the vector and the z -axis during the lifting portion of the rep. Notice the negative slope as the user's forearm completes half of the lifting phase. Continuing past this position towards the chin, however, increases the angle. These findings suggest that Barbell Curls, when executed in our reference frame, do not qualify as input to the above algorithm as the pattern they generate is unsupported. Thus, the observed behavior suggests transformations are needed to correct this issue. We apply a counter-clockwise rotation of 90° on the y -axis and observe the result in Figures 6a-6e. Notice that the transform causes the angle of the arms at rest to be significantly higher. Completing the first phase of the rep reduces the angle in approach to the chin and increases it when the trainee lowers the bar. Thus, we conclude that the transformation results in a signal that satisfies the algorithm's condition.

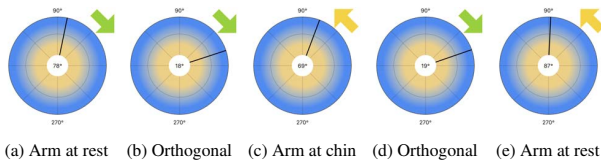


Figure 5: When a trainee performs the lifting portion of a repetition of a Barbell Curl, the slope of the angle between qpq^{-1} and the z -axis is initially negative (5a-5b) and later positive, 5c.

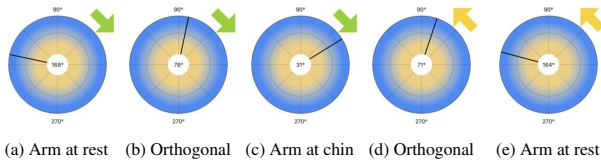


Figure 6: Rotating the y -axis by 90° of a single repetition of a Barbell Curl causes a negative slope when trainees lift the bar towards the chin and a positive slope when lowering to starting position.

5 RELATED WORK

Early work of visualizing quaternions explores twisting belts to portray how quaternions can represent orientation [13]. In recent work, however, most authors focus on displaying the different components onto a unit sphere and a unit circle [4, 12, 19, 27]. This strategy allows readers to develop an association between the scalar and imaginary parts, duality covering, and even the basics of transformations. Our effort is an extension of research in this area and builds on work by Mondol et al., who describe several interactive strategies for observing rotations with smart devices. The authors demonstrate how placing unit vectors onto the surface of a unit sphere helps visualize the orientation trace for human activities, such as eating or exercising. They also show the benefits of embedding a regular icosahedron within the unit sphere to help users distinguish trajectories of different activities more efficiently. We leverage these notable contributions to address several underexplored areas. First, polyhedrons with more vertices can better describe data in which rotations concentrate around a particular sphere region. Decoupling mesh granularity allows users to leverage a reference frame optimized for their specific data set. Second, since transformations are a vital component of inertial sensing, users need a mechanism to comprehend the outcome of rotating about an arbitrary axis. Lastly, with the scarcity of practical tools that describe quaternions, the entry barrier to inertial sensing can be high, requiring advanced mathematical knowledge.

In Section 4, we focus on applications of inertial sensing to fitness tracking with wearable devices. We briefly discuss related work in this area and how the ideas in this paper can complement the effort. Shen et al. leverage raw sensor readings, like acceleration, to detect cardio, weightlifting workouts, and rep counting [23]. This effort relates to Bhattacharya et al., who apply the context in deep learning for activity recognition [7]. Radhakrishnan et al. leverage inertial sensors mounted on weight equipment to classify repetition-level mistakes [21]. While inertial sensing at the hardware level is an appropriate strategy for kinematics, it is often noisy and sensitive. Since most modern devices use sensor fusion to construct quaternions, it opens an opportunity to represent the system state more accurately. The use of Euler angles is also present in this research area. Velloso et al. and O'Reilly et al. build models that use this technique to detect the validity of exercise execution [20, 26]. Elshafei et al. use similar representations to monitor muscle fatigue during Bicep Curls [9]. Balkhi et al. observe different extremities' pitch, roll, and yaw to build machine-learning models for activity recognition and rep counting [5]. While this mathematical method for orientation is intuitive, it suffers from singularities that the user must explicitly handle [2]. While not as popular, some work applies quaternions in the fitness domain. Guo et al. build a system for exercise classification and performance metric extraction [11]. Seuter et al. apply them to support vector machines to detect different types of movement, such as running [22]. However, the authors do not share a tool to prototype the findings of their study.

6 CONCLUSION

Quaternions are among the most efficient techniques for tracking orientation via inertial sensing. Despite their effectiveness in representing rotations, limited resources are available for visualizing quaternions in a meaningful way. This scarcity creates a conundrum where one must learn the concepts to visualize them, yet visualizing them is essential for understanding the concepts. To address this challenge, we build a tool that allows users to visualize quaternions, apply transformations, and reduce the dimensionality from four to one in real-time. With this tool, users can interactively study their systems' dynamic behavior without the need for repeated data collection, plotting, and external software. The methods and visualization tools presented in this work have a wide range of applications, making them valuable for researchers and practitioners alike.

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